

Project Title:**Waste Management Efficiency by Zone****Objective:**

The main objective of this project is to analyze waste generation, recycling efficiency, and landfill utilization across various Indian cities or districts. The goal is to identify which zones are performing efficiently in waste management and which require improvement, based on recycling rates, municipal efficiency scores, and cost factors.

Dataset Used:

Dataset Name: *Waste Management and Recycling in Indian Cities* (Kaggle)

Columns in Dataset:

- City/District
- Waste Type
- Waste Generated (Tons/Day)
- Recycling Rate (%)
- Population Density (People/km²)
- Municipal Efficiency Score (1–10)
- Disposal Method
- Cost of Waste Management (₹/Ton)
- Awareness Campaigns Count
- Landfill Name
- Landfill Location (Lat, Long)
- Landfill Capacity (Tons)
- Year

Data Period: Multiple years (as per dataset)

Data Preparation (Power Query):

1. **Loaded dataset** into Power BI and checked for data types and null values.
2. **Removed duplicates** and irrelevant columns.

- 3. **Renamed columns** for consistency and readability.
- 4. **Created new columns** where necessary, e.g., derived metrics like Landfill Utilization.
- 5. **Ensured numeric fields** (Waste Generated, Recycling Rate, Cost, etc.) were correctly formatted for calculations.

Data Modeling:

- A **Date Table (Year Table)** was created and related to the main Waste dataset through the “Year” column.
- This supports **time-based analysis** and enables DAX measures like YoY change and trends.
- Model relationships were verified using *Manage Relationships* in Power BI.

DAX Measures Created:

Measure Name	Formula / Description
Total Waste Generated	SUM(Waste[Waste Generated (Tons/Day)])
Average Recycling Rate (%)	AVERAGE(Waste[Recycling Rate (%)])
Average Efficiency Score	AVERAGE(Waste[Municipal Efficiency Score (1-10)])
Average Cost per Ton (₹)	AVERAGE(Waste[Cost of Waste Management (₹/Ton)])
Total Awareness Campaigns	SUM(Waste[Awareness Campaigns Count])
Landfill Utilization (%)	(SUM(Waste[Waste Generated (Tons/Day)]) / SUM(Waste[Landfill Capacity (Tons)])) * 100
Weighted Recycling Rate (%)	(SUMX(Waste, Waste[Waste Generated (Tons/Day)] * Waste[Recycling Rate (%)])) / SUM(Waste[Waste Generated (Tons/Day)])

Visualizations Used in Dashboard:

Visualization	Purpose
KPI Cards	To show key metrics: Total Waste, Avg. Recycling Rate, Avg. Efficiency, Avg. Cost, Total Campaigns
Clustered Bar Chart	Compare waste generation and recycling rate by City/District
Map Visualization	Show landfill locations and efficiency by region
Donut Chart	Display waste composition or disposal methods
Line Chart	Trend of waste generation and recycling over years
Table with Conditional Formatting	Display detailed metrics by city with color-based highlights

Conditional Formatting Applied:

- **Weighted Recycling Rate (%):** Green (High) → Red (Low) gradient to highlight performance.
 - **Landfill Utilization (%):** Red highlight for values above 80% to indicate overuse.
 - **Efficiency Score:** Darker shades for higher efficiency.
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Interactivity & Design Features:

- **Slicers:** Year, City/District, and Waste Type for flexible filtering.
 - **Tooltips:** Default hover tooltips used for clarity.
 - **No Drillthrough Page Needed:** All key insights and city-level breakdowns are integrated into the main dashboard.
 - **Clean Layout:** Single-page design for simplicity and quick insight extraction.
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Dashboard Title:

“Waste Management Efficiency by Zone”

Key Insight (One-Line Summary):

Zones with higher recycling rates and municipal efficiency scores consistently show reduced landfill pressure and optimized waste management costs.

Conclusion:

This project provided a clear view of how different Indian cities handle waste management. Insights revealed significant differences in efficiency, cost, and landfill usage among zones. Efficient cities tend to have stronger awareness programs and better recycling systems.

The Power BI dashboard helps policymakers and urban planners identify underperforming regions, optimize waste disposal methods, and plan capacity improvements for landfills.

Overall, the project successfully demonstrates the analytical workflow — from **data cleaning and modeling** to **visualization and storytelling** — using Power BI.